Week 6 DevOps Internship Report

Infrastructure as Code & Configuration Management

1. Infrastructure as Code (IaC)

Infrastructure as Code (IaC) is the practice of defining infrastructure through configuration files so that infrastructure can be created consistently and automatically instead of setting up every server manually. In the Week 6 task, IaC is represented by Terraform.

The main benefit is repeatability: the same configuration can be used to create a consistent environment and can be reviewed before changes are applied.

2. Terraform

Terraform is used to automate infrastructure provisioning. A Terraform configuration defines the desired infrastructure, and Terraform uses providers to communicate with the target platform. In this practical task, the Docker provider is used to provision a Docker image and container.

The practical Terraform workflow used in this task is: terraform init, terraform plan, terraform apply, and terraform destroy. The plan should be reviewed before applying changes.

3. Terraform State

Terraform state records information about resources managed by Terraform. Terraform uses the state to compare the current infrastructure with the desired configuration and determine what changes are required.

The Week 6 task requires understanding Terraform state and using Terraform state commands as part of the practical workflow.

4. Variables & Outputs

Terraform variables allow configuration values to be defined separately from resource definitions. In this task, variables are used for the Docker container name and host port.

Terraform outputs expose useful values after provisioning. The practical configuration outputs the provisioned container name and the web server URL.

5. Modules

Terraform modules are reusable collections of Terraform configuration. Modules help organize infrastructure configuration and make commonly used infrastructure patterns reusable and maintainable.

The Week 6 task requires the report to explain modules; a separate custom module was not required for the basic practical configuration described in the task.

6. Ansible

Ansible is used for configuration management. It configures and manages servers consistently after infrastructure has been provisioned.

For the Weekly Task, Ansible is used to install a package, create a user, and start a service through an Ansible playbook.

7. Inventory

An Ansible inventory defines the hosts that Ansible manages. The Weekly Task requires a basic inventory. The practical configuration uses localhost as the managed host with a local Ansible connection.

8. Playbooks

An Ansible playbook is a YAML file containing plays and tasks that describe the desired configuration of managed hosts. The Week 6 playbook contains tasks for installing a package, creating a user, installing a service package, and starting/enabling the service.

9. Terraform vs Ansible

Terraform: focuses on provisioning infrastructure. It describes infrastructure resources and manages their lifecycle.

Ansible: focuses on configuration management. It configures software, users, packages, and services on provisioned systems.

They can be used together: Terraform provisions the required infrastructure, and Ansible configures the provisioned server. This distinction is the central concept of the Week 6 Infrastructure as Code and Configuration Management topic.

Conclusion: Terraform and Ansible address different stages of infrastructure automation. Terraform provides repeatable infrastructure provisioning, while Ansible provides consistent configuration management. Together they support repeatable and maintainable DevOps environments.